

ORIGINAL ARTICLE



Sakuranetin exerts anticonvulsant effect in bicuculline-induced seizures

Wilson Vicente-Silva¹ | Francisca Rayanne Silva-Freitas¹ |
 José Ivo Araújo Beserra-Filho¹ | Gabriela Nascimento Cardoso¹ |
 Suellen Silva-Martins¹ | Tamires Alves Sarno¹ | Sara Pereira Silva¹ |
 Beatriz Soares-Silva¹ | José Ronaldo dos Santos² | Regina Helena da Silva³ |
 Carla Máximo Prado¹ | Anderson Keity Ueno⁴ | João Henrique Ghilardi Lago⁵ |
 Alessandra Mussi Ribeiro¹

¹Department of Biosciences, Federal University of São Paulo, Santos, São Paulo, Brazil

²Department of Biosciences, Federal University of Sergipe, Itabaiana, Sergipe, Brazil

³Department of Pharmacology, Federal University of São Paulo, São Paulo, São Paulo, Brazil

⁴Department of Biosciences, Federal University of São Paulo, Diadema, São Paulo, Brazil

⁵Center of Natural Sciences and Humanities, Federal University of ABC, Santo André, São Paulo, Brazil

Correspondence

Alessandra Mussi Ribeiro, Departamento de Biotecnologias, UNIFESP, Rua Silva Jardim 136, Edifício Central, CEP 11015-020 - Santos, SP, Brazil.
 Email: alessandra.ribeiro@unifesp.br

Funding information

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Grant/Award Number: 001; Conselho Nacional de Desenvolvimento Científico e Tecnológico, Grant/Award Number: 425694/2016-0; Fundação de Amparo à Pesquisa no Estado de São Paulo, Grant/Award Number: 2018/26609-5

Abstract

Epilepsy is a chronic neurological disorder characterized by an abnormal, spontaneous, and synchronized neuronal hyperactivity. Therapeutic approaches for controlling epileptic seizures are associated with pharmacoresistance and side effects burden. Previous studies reported that different natural products may have neuroprotector effects. Sakuranetin (SAK) is a flavanone with antiparasitic, anti-inflammatory, antimutagenic, antiallergic, and antioxidant activity. In the present work, the effect of SAK on seizures in a model of status epilepticus induced by bicuculline (BIC) in mice was evaluated. Male Swiss mice received an intracerebroventricular injection (i.c.v.) of SAK (1, 10, or 20 mg/kg—SAK1, SAK10, or SAK20). Firstly, animals were evaluated in the open field (OF; 20 min), afterwards in the elevated plus maze (EPM) test (5 min). Next, 30 min prior the administration of BIC (1 mg/kg), mice received an injection of SAK (1 or 10 mg/kg, i. c.v.) and were observed in the OF (20 min) for seizures assessment. After behavioral procedures, immunohistochemical analysis of c-Fos was performed. Our main results showed that the lowest doses of SAK (1 and 10 mg/kg) increased the total distance traveled in the OF, moreover protected against seizures and death on the BIC-induced seizures model. Furthermore, SAK treatment reduced neuronal activity on the dentate gyrus of the BIC-treated animals. Taken together, our results suggest an anticonvulsant effect of SAK, which could be used for the development of anticonvulsants based on natural products from herbal source.

KEYWORDS

epilepsy, flavonoids, hippocampus, neuroprotection, seizures

Abbreviations: AEDs, antiepileptic drugs; ANOVA, analysis of variance; BIC, bicuculline; CD₃OD, deuterated methanol; CEDEME, Center of the Development of Experimental Models for Medicine and Biology; CH₂Cl₂, dichloromethane; DG, dentate gyrus; DMSO, dimethylsulfoxide; DZP, diazepam; EPM, elevated plus maze; EtOAc, ethyl acetate; EtOH, ethanol; GABA, gamma-aminobutyric acid; HPLC, high-performance liquid chromatography; HRESIMS, high-resolution electrospray ionization mass spectrometry; i.c.v., intracerebroventricular; i.p., intraperitoneal; JNK, c-Jun N-terminal protein kinase; NMR, nuclear magnetic resonance; OF, open field; s.c., subcutaneous; SAK, sakuranetin; SAL, saline; STAT1, signal transducer and activator of transcription 1; TLC, thin-layer chromatography.

1 | INTRODUCTION

Epilepsy is a chronic brain disorder characterized by an abnormal, spontaneous, and coordinated neuronal hyperactivity leading to deleterious neuronal changes and negative social impact [1–3]. It is estimated that about 70 million people worldwide are diagnosed with epilepsy [4, 5]. Epileptic seizures are result from the imbalance between the excitatory and inhibitory neurotransmission, synaptic recombination, and glial cell proliferation [6, 7]. Disturbances in the balance between excitatory/inhibitory may lead to dysfunctional signaling resulting in behavioral deficits [8]. Moreover, evidence shows an important role of neuroinflammation and oxidative stress in the onset of *status epilepticus* [9, 10].

Mesial temporal lobe epilepsy (MTLE) associated with hippocampal sclerosis is the most common pattern of epilepsy. MTLE is characterized by the neuronal loss in the CA1, CA3, and CA4 areas of the hippocampus (HIP). CA2 and dentate gyrus (DG) are resistant to cell loss [11]. Moreover, status epilepticus cause alterations in neurogenesis of the DG granular zone and apoptosis in pyramidal cells of the CA1 and CA3 areas [12]. Hippocampal resection is one of the most common treatment approaches in patient's refractory to pharmacological therapy [13, 14].

Antiepileptic drugs (AEDs) modulate GABAergic and glutamatergic neurotransmission or voltage-dependent ion channels [15]. Despite several classes of AEDs commercially available, about 30% of the patients develop pharmacoresistance [16, 17]. Furthermore, chronic treatment with AEDs induces side effects that contribute to the decline in the quality of life [18, 19]. In view of this, novel therapeutic approaches to the treatment of epilepsy are needed. Compounds obtained from plant extracts, such as flavonoids, demonstrate therapeutic properties on several animal models of seizures reinforcing that these substances could be a potential source of model drug candidate entities for the treatment of epilepsy [20].

Sakuranetin (5,4'-dihydroxy-7-methoxyflavanone; SAK) is a flavonoid present in different species of plants [7, 21, 22] with several pharmacological properties such as antifungal [23], antitumoral [24], antiparasitic [25], antimicrobial [26, 27], and antioxidant [28, 29]. In addition, SAK showed anti-inflammatory activity by inhibiting of the expression of NF- κ B and pro-inflammatory cytokines in an animal model of lung disease and in vitro [30–32]. Furthermore, SAK modulates the activation of the c-Jun N-terminal protein kinase (JNK), mitogen-activated protein kinase (p38), and signal transducing and transcription-activating proteins (STAT) [30, 33]. Interestingly, this flavonoid also showed protective effects on neurons through antioxidation and anti-inflammatory mechanisms [7]. In this context, compounds with anti-inflammatory and antioxidant properties may have anti-epileptic effect [34]. Thus, we hypothesized that SAK

may have an anticonvulsant effect because of its anti-inflammatory and antioxidant activities. The aim of this study was to evaluate the anticonvulsant potential of SAK on the seizure model induced by bicuculline (BIC) and the effects of treatment on the neuronal activity of different hippocampal areas (CA1, CA3, and DG).

2 | MATERIAL AND METHODS

2.1 | General experimental procedures

Chromatographic procedures were performed using silica gel 60 (Merck, 230–400 mesh) for column and silica gel 60 PF254 (Merck) for analytical thin-layer chromatography (TLC) (0.25 mm). Nuclear magnetic resonance (NMR) spectra were recorded in a Bruker DPX-300 spectrometer operating at 300 MHz (^1H) and 75 MHz (^{13}C) using CD_3OD (deuterated methanol; Aldrich) as solvent. High-resolution electrospray ionization mass spectrometry (HRESIMS) was measured on a Bruker Daltonics MicroTOF QII spectrometer, at positive ionization mode.

Prior to the start of experimental procedures, mice were handled carefully (5 min/5 days). In order to protect the animals from the aversion to light, the behavioral tests were carried out under low light conditions (30 lux). Each behavioral apparatus was cleaned (alcohol 5%) after each behavioral trial. The experimental sessions were recorded and examined by the video-tracking computer software (Any-maze[®], Stoelting, USA), excluding the seizures evaluation that were analyzed manually. A blind observer carried out the behavioral assessment performed manually.

2.2 | Plant material

Baccharis retusa DC. aerial parts were collected in Campos do Jordão, SP, Brazil, in October 2016 from the same specimen studied previously [35].

2.3 | Extraction of SAK

Dried and powdered aerial parts of *B. retusa* (500 g) were defatted using hexane (6 $\times$ 500 ml) and then exhaustively extracted using ethanol (EtOH) (10 $\times$ 500 ml) at room temperature. Crude EtOH extract (41 g) was resuspended in EtOH:H₂O 2:1 and successively extracted using hexane (8 g) and CH_2Cl_2 (dichloromethane; 14 g). Part of the CH_2Cl_2 phase (10 g) was subjected to column chromatography over silica gel eluted with increasing amounts of ethyl acetate (EtOAc) in hexane to afford four groups (A–D). SAK was obtained, in pure form as evidenced by high-performance liquid chromatography (HPLC), from Group B (844 mg).

2.3.1 | 5,4'-Dihydroxy-7-methoxyflavanone (SAK)

White amorphous solid. ^1H NMR (300 MHz, CD_3OD) δ_{H} 7.26 (d, $J = 8.5$ Hz, H-2'/H-6'), 6.83 (d, $J = 8.5$ Hz, H-3'/H-5'), 6.01 (s, H-6/H-8), 5.32 (dd, $J = 13.0$ and 3.0 Hz, H-2), 3.77 (s, OCH_3 -7), 3.08 (dd, $J = 17.2$ and 13.0 Hz, H-3a), 2.73 (dd, $J = 17.2$ and 3.0 Hz, H-3b). ^{13}C NMR (75 MHz, CD_3OD) δ_{C} 196.5 (C-4), 168.0 (C-4'), 163.6 (C-7), 163.0 (C-5), 157.4 (C-9), 129.0 (C-1'), 127.7 (C-2'/C-6'), 127.6 (C-6), 115.3 (C-3'/C-5'), 102.8 (C-10), 93.9 (C-8), 79.1 (C-2), 55.3 (OCH_3), 42.8 (C-3). HRESIMS m/z 287.0916 $[\text{M} + \text{H}]^+$.

2.4 | Animals

The study was performed on male Swiss mice (30–60 g, 3–4 months old, $n = 60$) that were acquired from the Center of the Development of Experimental Models for Medicine and Biology (CEDEME—Federal University of São Paulo). Animals were randomly housed in groups (five to six animals per plastic cages, $30 \times 20 \times 16$ cm) under standard conditions of acoustic isolation and controlled airflow and temperature ($22\text{--}24^\circ\text{C}$), with a light–dark cycle 12/12 h (lights on 7 a.m.). Water and food were available ad libitum. All the experiments were conducted in accordance to the Brazilian law for the use of animals in research (Law Number 11.794) and approved by the Ethics Committee from the Federal University of São Paulo (UNIFESP; Protocol No. 5705251117). All efforts were made to diminish animal discomfort or distress.

2.5 | Surgery

Mice were submitted to surgical method for the implantation of a guide cannula (10 mm, lateral ventricle). Firstly, animals were anesthetized with an intraperitoneal (i.p.) injection of a mixture of ketamine (100 mg/kg), xylazine (10 mg/kg), fentanyl citrate (0.1 mg/kg), and acepromazine (1 mg/kg). Then, mice received a subcutaneous (s.c.) injection of pentabiotic benzylpenicillin benzathine (antibiotic, 50 000 UI/kg) and anti-inflammatory. Afterwards, mice were placed in the stereotaxic frame (Insight, Brazil) and the skull was exposed for the cannula implantation. The stereotaxic coordinates used in this procedure were as follows: anterior–posterior = 0.2 mm; meso-lateral = 1.0 mm on the left side; and dorsoventrally: 2.2 mm, from the bregma [36]. Then, the guide cannula was fixed to the skull with dental acrylic and sealed with stainless steel wire (to avoid obstructions). Animals received Meloxicam[®] (anti-inflammatory drug, 2 mg/kg; s.c.) 24 and 48 h after the surgery and were monitored for 5 consecutive days (recovery time) prior to behavioral procedures.

2.6 | Experimental groups and behavioral assessment

Diazepam (DZP) and BIC were diluted in saline (SAL) solution and administered intraperitoneally (i.p.). SAK was diluted in dimethylsulfoxide (DMSO) (Sigma-Aldrich; #14340). Thus, SAL and DMSO groups were control groups. All substances were injected intracerebroventricularly (i.c.v.), at a final volume of 1 μl through an infusion pump (Insight) attached to a glass microsyringe (10 μl ; Hamilton Co., USA). The device was attached to an injection needle (11 mm) that was inserted into the cannula (10 mm). The experimental design is shown in Figure 1.

Animals were allocated to one of six groups: CTR (0.9% SAL, i.c.v., $n = 7$); DMSO (DMSO i.c.v., $n = 8$); SAK1 (SAK 1 mg/kg i.c.v., $n = 9$); SAK10 (SAK 10 mg/kg i.c.v., $n = 7$); SAK20 (SAK 20 mg/kg i.c.v., $n = 7$); and DZP (DZP 2 mg/kg i.p., $n = 7$).

After SAK, DMSO, or DZP treatment, mice were immediately placed to the circular arena (open field [OF] test, 20 min) carried out in an apparatus (40 cm $\times$ 50 cm) made of wood and painted black, positioned in a lighted lab room. Total distance traveled, time spent in the exploration of the inner zone of apparatus, time of grooming, and time of immobility were considered to the behavioral evaluation in the OF. The immobility time was quantified from 2 s without movement or changes in the animal's position.

After OF test, mice were evaluated in the elevated plus maze (EPM) test (5 min) that consisted in a cross-shaped wooden equipment made with two enclosed arms (EA; 27.5 cm $\times$ 6.5 cm $\times$ 18 cm) opposite to two open arms (OA; 27.5 cm $\times$ 6.5 cm). Time spent in the OA and EA and risk assessment behaviors (protected head dipping [PHD]: when the mouse performed from the center of apparatus a scanning over the sides of the maze downward; stretched attend posture [SAP]: when the animal stretching and contraction of body to its original position without locomotion) were evaluated. Then, after behavioral tests, each animal was euthanized by overdose of anesthetic.

2.6.1 | Potential anticonvulsant effect of SAK in mice submitted to the seizure model induced by BIC

Animals were allocated to the following groups: SAL-BIC (0.9% SAL i.c.v. + BIC 1 mg/kg i.c.v., $n = 5$), DMSO-BIC (DMSO i.c.v. + BIC 1 mg/kg i.c.v., $n = 6$), SAK1-BIC (SAK 1 mg/kg i.c.v. + BIC 1 mg/kg i.c.v., $n = 8$), and SAK10-BIC (SAK 10 mg/kg i.c.v. + BIC 1 mg/kg i.c.v., $n = 6$).

SAL, DMSO, or SAK was administered 30 min prior the BIC injection. Then, mice were immediately placed to the OF (20 min) for seizures behavior assessment.

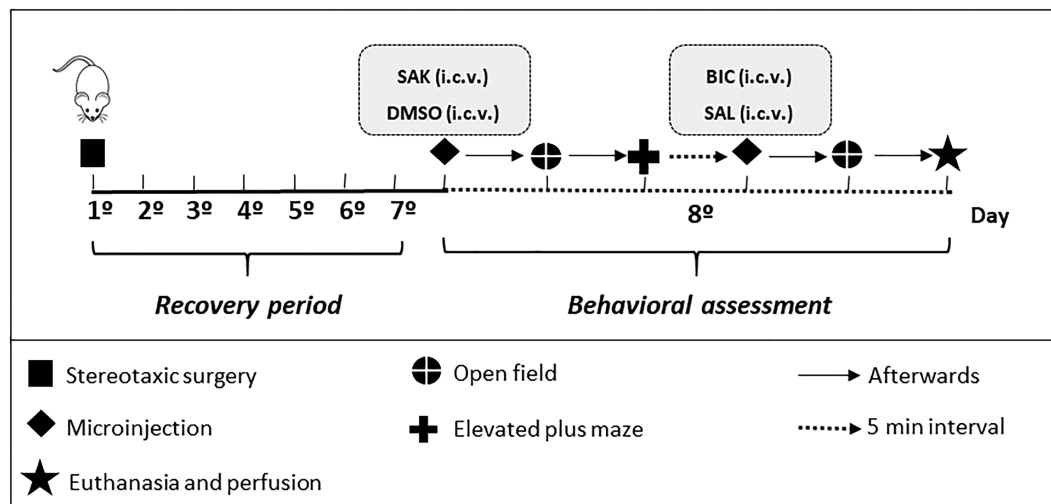


FIGURE 1 Schematic representation of the experimental design

The animals were euthanized and brains collected 1 h after BIC administration to assess c-Fos immunoreactivity. For this analysis, c-Fos protein maximum expression interval [37] and animal welfare were considered. Microinjection sites were checked by administration of methylene blue (1 μ l) into the guide cannula. The administration of this substance allowed the visualization of the correct site using a microscope (bright field). Only mice with correct microinjection sites were considered in our study.

The severity of seizures was analyzed by modified Racine scale [38] (adapted by Nôga et al. [39]): (1) myoclonic jerks of contralateral paw; (2) mild paw clonus lasting at least 5 s; (3) severe paw clonus lasting at least 15 s; (4) rearing in addition to severe paw clonus; and (5) rearing and falling in addition to severe paw clonus. Latency to the onset of tonic-clonic seizures (time between outset of behavioral assessment and first episode of score-5 seizure), latency to the death (time interval until death), and percentage of protection against tonic-clonic seizures (number of animals that develop tonic-clonic seizure/number of animals in the group $\times$ 100) and survival (number of animals that survived/number of animals in the group $\times$ 100) were also considered.

2.6.2 | Perfusion and immunohistochemistry

After behavioral tests, each animal was quickly euthanized with an overdose of anesthetic (sodium thiopental, 120 mg/kg, i.p.) and previous injection of lidocaine 2% (10 mg/ml, i.p.). Afterwards, mice were perfused transcardially. The brain was removed from the skull, postfixed in the same fixative solution of paraformaldehyde 4% (PFA 4%) for 24 h. After, the sucrose

protected brain was drained and frozen at -16°C . The brain was serially cut in 50 μm sections and placed sequentially in five compartments (one section per compartment) with antifreeze solution.

For the immunohistochemistry assessment, brain sections were initially washed four times (5 min each) in 0.1 M phosphate buffered saline (PBS) and exposed to 0.0003% H_2O_2 solution (20 min) to block endogenous peroxidase activity. Thus, samples were incubated in fresh skimmed milk 3% diluted in 0.1 M PBS for 15 min. Subsequently, the sections were incubated with the primary antibody for c-Fos (1:2000, rabbit anti-c-Fos polyclonal antibody, mAb#2250 Cell Signal, USA) [40] for 18 h and then bathed four times (5 min each) in 0.1 M PBS. After, the cuts were incubated in the secondary antibody (1:1000, goat biotinylated anti-rabbit IgG, #BA-1000, Vector Labs, USA) for 2 h. Then, they were again submitted to the bath in 0.1 M PBS (four times of 5 min) and incubated with the avidin-biotin complex (ABC kit, Vector Labs) for 2 h. Then, the sections were bathed in 0.1 M PBS (four times of 5 min) and incubated in 0.05% solution of 3,3'-diaminobenzidine (DAB; Sigma Company, USA) for 2 min to chromogen deposition. When an adequate level of color was reached, the cuts were subjected to an alcohol dehydration battery 70%, 95%, and 100% (two times) and xylene (two times), for 5 min each. After this stage, the cuts were mounted on slides and covered with inclusion media (Entellan, Merck) and mounted with coverslips.

Stereological counting of cells labeled for c-Fos was performed using an optical microscope (Nikon Eclipse Ni-E; 20 $\times$, Japan), attached with a digital camera (Motic 5.0) under bright-field illumination. The boundaries of each region of interest, pyramidal cells (CA1 and CA3), and DG granular zone were delimited according to the brain atlas of mouse [36]. We analyzed

these areas due to neuroanatomical alterations described previously. The following hippocampal areas were analyzed from -1.70 to -2.06 mm of the bregma. Cell counting was performed with $20\times$ objective lenses, and the density of positive cells was counted using the Image-Pro Plus Software (Media Cybernetics, USA). Cells were considered c-Fos immunoreactive if their nuclei contained punctate dark brown immunolabeling using constant minimum and maximum optical density (OD) and objective size criteria. Three or four sections were analyzed *per area*. The mean value of the sections *per area* analyzed was considered for each animal. Values were normalized by the mean value of BIC group. The cannulation site and hippocampal areas analyzed are shown in Figure 2.

2.7 | Statistical analysis

One-way analysis of variance (ANOVA) was used to analyze the OF, EPM, and seizure evaluation data. Differences between dependent factors were evaluated by Bonferroni's post hoc test. Fisher's exact test was used to analyze percentage of protection and survival. Results were expressed as mean $\pm$ SEM (standard error of mean) and a $P < 0.05$ was considered significant.

3 | RESULTS

3.1 | Chemical characterization of SAK

NMR (^1H and ^{13}C) and HRESIMS data of isolated compound from aerial parts of *B. retusa* were compared with those reported in the literature [35], allowing the identification of SAK (Figure 3), in 99% of purity as indicated by HPLC.

3.2 | Behavioral assessment

We found an effect of both SK and DZP treatments in the total distance traveled [$F_{(5,44)} = 4.1$, $P < 0.05$], time in the inner zone [$F_{(5,44)} = 6.6$, $P < 0.05$], time of grooming [$F_{(5,45)} = 4.9$, $P < 0.05$], and time of immobility [$F_{(5,45)} = 33.9$, $P < 0.05$] evaluated by OF test. Bonferroni's post hoc test showed that SAK1 and SAK10 groups increased the traveled distance compared to SAL group ($P < 0.05$), suggesting a hyperlocomotion. On the other hand, DZP-treated mice had less time spent in the inner zone, decreased time of grooming behavior, and increased time of immobility compared to SAL group ($P < 0.05$; Figure 4).

Regarding EPM test, we observed treatment effect for time in the OA [$F_{(5,44)} = 4.9$, $P < 0.05$], EA [$F_{(5,44)} = 3.5$, $P < 0.05$], PHD [$F_{(5,44)} = 2.9$, $P < 0.05$], and SAP [$F_{(5,44)} = 3.2$, $P < 0.05$]. The treatment with SAK10 increased the EA compared to DMSO group ($P < 0.05$). Moreover, SAK10 and SAK20 increased the time spent in the OA compared to DMSO and DZP groups ($P < 0.05$). On the risk assessment, only the SAK20 treatment increased the time of PHD and SAP behavior compared to DZP group ($P < 0.05$; Figure 5).

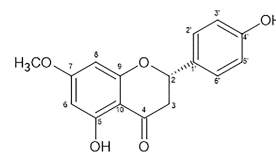


FIGURE 3 Chemical structure of sakuranetin (SAK), isolated from aerial parts of *Baccharis retusa*

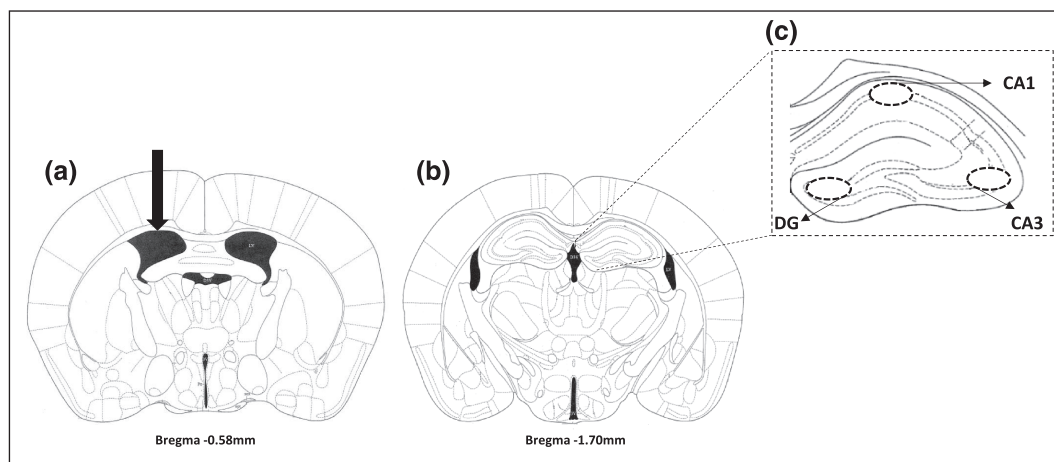


FIGURE 2 Schematic drawing illustrating (a) microinjection site (arrow) for mice injected with saline, bicuculline, or SAK [36]; (b, c) hippocampal areas investigated [36]. Cycles highlight the analyzed areas

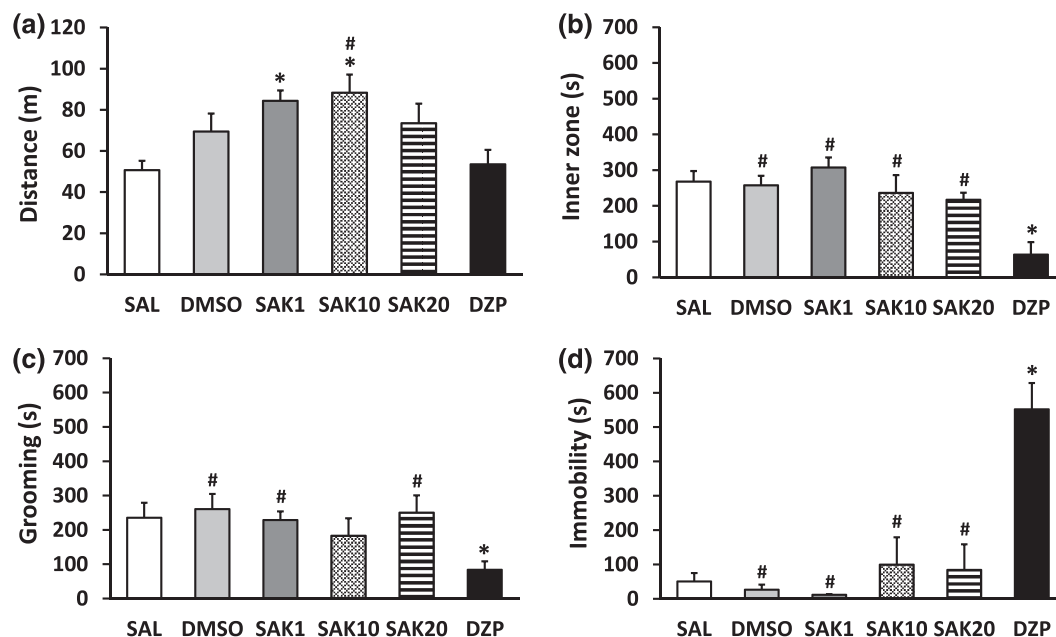


FIGURE 4 Effects of pre-training microinjection of sakuranetin (SAK; 1, 10, or 20 mg/kg) or diazepam (DZP; 2 mg/kg) on exploratory behavior in the OF test in mice. (a) Total distance traveled; (b) time spent in the inner zone; (c) time of grooming; (d) time of immobility. Data are expressed in mean ± SEM. * $P < 0.05$ compared to SAL; # $P < 0.05$ compared to DZP. One-way ANOVA followed by Bonferroni's test

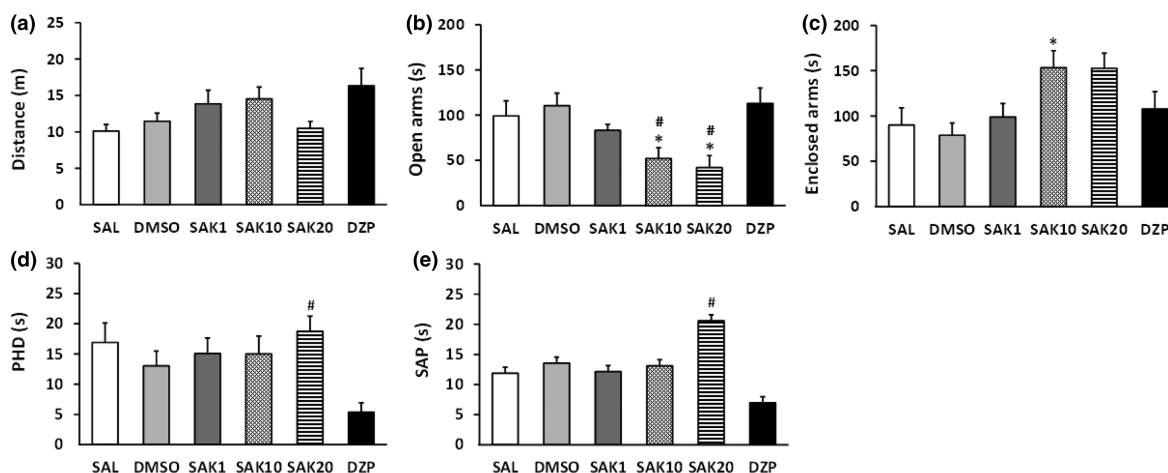


FIGURE 5 Effects of the administration of sakuranetin (SAK; 1, 10, or 20 mg/kg) or diazepam (DZP; 2 mg/kg) on anxiety-like behavior in the EPM in mice. (a) Total distance traveled; (b) time spent in the open arms (OA); (c) time spent in the enclosed arms (EA); (d) time of protected head dipping (PHD); (e) time of stretched attend posture (SAP). Data are expressed in mean ± SEM. * $P < 0.05$ compared to DMSO; # $P < 0.05$ compared to DZP. One-way ANOVA followed by Bonferroni's test

3.3 | Anticonvulsant behavior assessment

As expected, mice that received vehicle previously to BIC treatment demonstrated maximum severity of seizures (Table 1). On the other hand, the treatment with SAK showed anticonvulsant effect on this model. One-way ANOVA indicated effect of SAK for latency to onset of seizure [$F_{(3,24)} = 3.7$, $P < 0.05$] and death [$F_{(3,24)} = 2.6$, $P < 0.05$]. Bonferroni's post hoc analysis revealed that SAK10 treatment increased the latency of

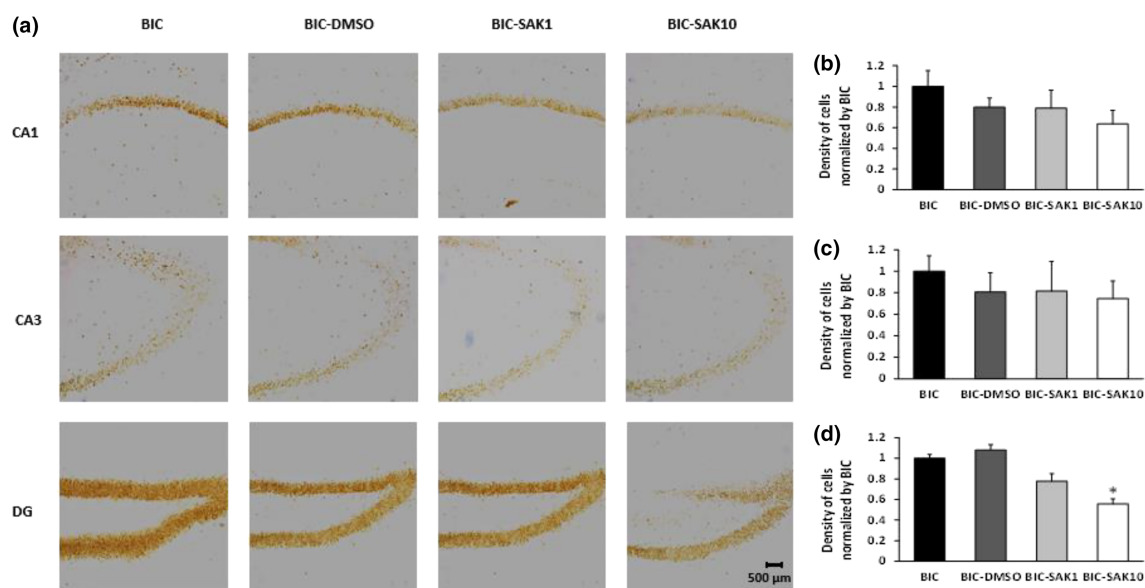
first seizure and death compared with SAL group ($P < 0.05$). Moreover, Fisher's exact test revealed that SAK treatment (10 mg/kg) protected the animals of seizures and mortality ($P < 0.05$, Table 1).

3.4 | Immunohistochemistry assessment of c-Fos

Regarding c-Fos cells immunoreactivity, one-way ANOVA indicated treatment effect in the hippocampal

TABLE 1 Effect of administration of SAK (1 or 10 mg/kg) and BIC (1 mg/kg) on mice submitted to the BIC-induced seizure model

	BIC	BIC-DMSO	BIC-SAK1	BIC-SAK10
Racine score ratio	5	5	2.5	0
Mortality	4/5	6/6	4/8	0/6
Latency of first seizure (s)	135.6 ± 48.6	106.5 ± 42.4	53.0 ± 15.0	0*
Latency to death (s)	741.0 ± 196.9	862.5 ± 125.9	962.3 ± 93.0	1200*
% Seizures protection	0%	0%	25%	100%
% Mortality protection	0%	0%	50%	100%

* $P < 0.05$ compared to BIC group.**FIGURE 6** Effects of treatment of sakuranetin (SAK; 1 or 10 mg/kg) and bicuculline (BIC; 1 mg/kg) on the c-Fos-immunoreactivity evaluation (bregma: from -1.70 to -2.06 mm [35]). (a) Representative photomicrographs of coronal sections in the hippocampus (scale bar: $500\ \mu\text{m}$); (b) density of c-Fos cells in CA1 area; (c) density of c-Fos cells in CA3 area; (d) density of c-Fos cells in the DG granular zone. All were normalized by BIC value. * $P < 0.05$ compared to BIC. One-way ANOVA followed by Bonferroni's test

formation [$F_{(3,15)} = 28.3$, $P < 0.05$]. Bonferroni's post hoc test revealed that the treatment with SAK10 decreased c-Fos cells immunoreactive in the DG, but not in other areas ($P < 0.05$; Figure 6).

4 | DISCUSSION

Previous studies showed pharmacological properties of SAK in different experimental models, such as antioxidant and anti-inflammatory properties that can be useful in epilepsy. In the first set of experiments, it was observed the effect of SAK treatment in different doses in the OF and EPM tests. Thereafter, SAK was evaluated on experimental model for convulsive seizures induced by BIC. Our results showed the SAK in low doses exerts potent anticonvulsant effect reducing the mortality.

The use of different natural products consists in an important strategy for development of potential candidates to treat disorders such as anxiety and depression [21, 41]. Natural products such as daidzein, an isoflavone, ameliorate mental disorders reducing side effects when compared to the conventional pharmacological treatment [41]. Evidence shows that some flavonoids have a neuroprotective potential on mental disorders [42–44] since these substances are able to cross the blood–brain barrier and show affinity with the GABA_A receptor, which is related to its sedative, anxiolytic, and anticonvulsant effects [45, 46]. SAK can cross biological membranes by passive diffusion [47]. Herein, SAK was microinjected intracerebrally to avoid a possible first-pass effect through the liver. In our study, mice treated with lower doses of SAK (1 and 10 mg/kg) increased the distance traveled in the OF suggesting a hyperlocomotor activity, however, at the highest dose

(20 mg/kg) did not observe locomotor alteration. Alterations in the GABAergic neurotransmission affect the locomotor activity in rodents. The inhibition of GABAergic signaling in the nucleus accumbens, for example, induced hyperlocomotion in rats [48].

Interestingly, previous studies showed that flavonoids have anxiolytic effects, for example, naringenin, a biosynthetic precursor of SAK with antioxidant activity, showed anxiolytic effect in rats [49]. Other flavonoids as quercetin, luteolin, and chrysin also demonstrate this effect in rodents [50–52]. Our findings showed that SAK treatment in the higher dose decreased the time spent in the OA and increased the time of PHD and SAP behavior compared to DZP group on the EPM, suggesting an anxiogenic-like behavior. However, in our study, DZP treatment provoked a motor alteration indicating sedative effect. Thus, novel studies are necessary to elucidate the behavioral alterations induced by SAK treatment in mice.

Upon completion the behavioral analysis in the OF and EPM, the anticonvulsant potential of SAK was evaluated. As mentioned, SAK demonstrated to be a promising strategy to the management of central nervous system (CNS) disorders. SAK treatment prevented learning and memory impairment in an Alzheimer's disease animal model [7]. However, the effects of SAK in an animal model of epilepsy remained unknown.

As mentioned before, epileptic seizures are caused by the imbalance in the inhibitory neurotransmission. Gamma-aminobutyric acid (GABA) is the major inhibitory neurotransmitter in the brain, in turn decrease of release this neurotransmitter in the synaptic cleft may induce the onset of seizures [53]. Moreover, the enhancement of the GABAergic neurotransmission is related to seizure blockade [54]. In view of this, BIC is an antagonist of receptor $GABA_A$ that disrupt the GABAergic neurotransmission [55] inducing tonic-clonic seizures in rodents [56]. BIC-induced seizures model provokes neuronal alterations in rodents similar to the electroencephalographical (EEG) discharge on the epilepsy patients [57, 58]. Moreover, this model has been showing effectiveness in the evaluation of novel candidates to be considered available for the treatment of epilepsy [39, 59–69]. Thus, we used this model to investigate the possible anticonvulsant effect of SAK in mice. In our study, intracerebral administration of SAK (10 mg/kg) prevented mice from developing BIC-induced tonic-clonic seizures and protected animals from death. Moreover, the lowest dose (1 mg/kg) of SAK protected 25% of the animals from seizures and 50% from death, also suggesting an anticonvulsant activity.

HIP plays an important role in the epileptogenic process of MTLE patients. Previous studies showed a decrease in the volume of HIP, especially in the focal epilepsy [70, 71]. Hippocampal sclerosis is

accompanied by gliosis, synaptic disorganization, and alterations of cellular structures within the subregions of HIP [72]. As mentioned, hippocampal resection is commonly used as treatment alternative in drug-resistant patients [73]. In our study, SAK10 treatment reduced the immunoreactivity of c-Fos in the DG of the animals, suggesting a reduced neuronal activity in this area. No differences were observed in CA1 and CA3 areas with BIC or SAK treatment. HIP is sensitive to stress and its subareas respond differently to stressful stimuli [74]. Thus, further studies are required to clarify this issue.

DG is composed by glutamatergic neurons called granule cells that establish excitatory connections with the HIP through entorhinal cortex [75]. DG plays a critical role in the behavior and cognition including mood, anxiety, and spatial memory [76], as well in CNS disorders such as epilepsy [77]. The persistent activation of granule cells increases the levels of glutamate inducing hippocampal neurons toxicity [78]. On the other hand, Botterill et al. showed that the inhibition of granule cells in the DG reduced seizure behavior and hippocampal injury [79]. Moreover, the decrease in neuronal activity in the DG cells may be a strategy for the temporal lobe epilepsy management [80]. Thus, the reduced neuronal activity observed in the DG of the animals in our study may be related with the anticonvulsant effect of SAK.

As mentioned, SAK has antioxidant and anti-inflammatory properties [7] and flavonoids have anticonvulsant effect mainly on GABA-related benzodiazepine sites [46, 81, 82]. The model of BIC-induced seizures is based on blocking GABAergic neurotransmission; therefore, we consider that SAK may also act on GABA receptor. Further studies are needed to clarify its mechanism(s) of action.

5 | CONCLUSION

Taken together, our results showed that SAK treatment protected animals from the seizures induced by BIC injection and it is associated to a reduction in the immunoreactivity of c-Fos in the DG. Although further studies are needed to clarify the action mechanism of the anticonvulsant effects of SAK, this compound seems to be a potent candidate as an anticonvulsive drug.

ACKNOWLEDGMENTS

This research was supported by fellowships from Fundação de Amparo à Pesquisa no Estado de São Paulo (FAPESP; Grant #2018/26609-5), Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq; Grant #425694/2016-0), and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES; Finance Code 001).

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

WVS, FRSF, and AMR designed the study. WVS, FRSF, GNS, SSM, TAS, BSS, and AKU performed the experiments and collected the data. WVS, FRSF, JIABF, and AMR analyzed the data and wrote the manuscript. JRS, RHS, CMP, JHGL, and AMR contributed with theoretical discussions and technical insights. WVS, FRSF, JIABF, JRS, RHS, CMP, JHGL, and AMR contributed in analysis and writing and revised the final manuscript. All the authors have read the final manuscript and approved the submission.

DATA AVAILABILITY STATEMENT

The datasets used during the current study are available from the corresponding author on reasonable request.

ORCID

Alessandra Mussi Ribeiro  <https://orcid.org/0000-0002-7697-5766>

REFERENCES

- Ding K, Gupta PK, Diaz-Arrastia R. Epilepsy after traumatic brain injury. *Transl Res Trauma Brain Inj*. 2016;299-313.
- Gelfuso EA, Liberato JL, Cunha AOS, et al. Parawixin2, a novel non-selective GABA uptake inhibitor from *Parawixia bistrata* spider venom, inhibits pentylenetetrazole-induced chemical kindling in rats. *Neurosci Lett*. 2013;543:12-16. doi:10.1016/j.neulet.2013.02.074
- Rosenberg EC, Tsien RW, Whalley BJ, Devinsky O. Cannabinoids and epilepsy. *Neurotherapeutics*. 2015;12(4):747-768. doi:10.1007/s13311-015-0375-5
- Ngugi AK, Kariuki SM, Bottomley C, Kleinschmidt I, Sander JW, Newton CR. Incidence of epilepsy: a systematic review and meta-analysis. *Neurology*. 2011;77(10):1005-1012. doi:10.1212/WNL.0b013e31822cfc90
- Thijs RD, Surges R, O'Brien TJ, Sander JW. Epilepsy in adults. *Lancet*. 2019;393(10172):689-701. doi:10.1016/S0140-6736(18)32596-0
- DiNuzzo M, Mangia S, Maraviglia B, Giove F. Physiological bases of the K⁺ and the glutamate/GABA hypotheses of epilepsy. *Epilepsy Res*. 2014;108(6):995-1012. doi:10.1016/j.epilepsyres.2014.04.001
- Li C, Hu C, Wang R, et al. Protective effect of sakuranetin in brain cells of dementia model rats. *Cell Mol Biol*. 2019;65(5):54-58. doi:10.14715/cmb/2019.65.5.9
- Sears S, Hewett S. Influence of glutamate and GABA transport on brain excitatory/inhibitory balance. *Exp Biol Med*. 2021;246(9):1069-1083. doi:10.1177/1535370221989263
- Ambrogini P, Torquato P, Bartolini D, et al. Excitotoxicity, neuroinflammation and oxidant stress as molecular bases of epileptogenesis and epilepsy-derived neurodegeneration: the role of vitamin E. *BBA-Mol Basis Dis*. 2019;1865(6):1098-1112. doi:10.1016/j.bbadis.2019.01.026
- Terrone Gaetano, Balosso Silvia, Pauletti Alberto, Ravizza Teresa, Vezzani Annamaria. Inflammation and reactive oxygen species as disease modifiers in epilepsy. *Neuropharmacology*. 2020;167(107742):1-10. doi:10.1016/j.neuropharm.2019.107742
- Nayak CS, Bandyopadhyay S. Mesial temporal lobe epilepsy. *StatPearls*. 2021;(26):1-10.
- Zhu L, Dai S, Lu D, Xu P, Chen L. Role of NDEL1 and VEGF/VEGFR-2 in mouse hippocampus after status epilepticus. *Am Soc Neurochem*. 2020;12:1-9. doi:10.1177/1759091420926836
- Cramer SW, McGovern RA, Wang SG, Chen CC, Park MC. Resective epilepsy surgery: assessment of randomized controlled trials. *Neurosurg Rev*. 2021;44(4):2059-2067. doi:10.1007/s10143-020-01432-x
- Herlin B, Adam C, Odile M, et al. Temporal pole epilepsy surgery—sparing the hippocampus. *Epilepsia*. 2020:1-7.
- Rogawski MA, Löscher W, Rho JM. Mechanisms of action of antiepileptic drugs and the ketogenic diet. *Cold Spring Harb Perspect Med*. 2016;6(5):1-28. doi:10.1101/cshperspect.a022780
- Holtkamp D, Opitz T, Niespodziany I, Wolff C, Beck H. Activity of the anticonvulsant lacosamide in experimental and human epilepsy via selective effects on slow Na⁺ channel inactivation. *Epilepsia*. 2017;58(1):27-41. doi:10.1111/epi.13602
- Kwan P, Arzimanoglou A, Berg AT, et al. Definition of drug resistant epilepsy: consensus proposal by the ad hoc Task Force of the ILAE Commission on Therapeutic Strategies. *Epilepsia*. 2010;51(6):1069-1077. doi:10.1111/j.1528-1167.2009.02397.x
- Ghosh S, Sinha JK, Khan T, et al. Pharmacological and therapeutic approaches in the treatment of epilepsy. *Biomedicine*. 2021;9(470):1-14. doi:10.3390/biomedicine9050470
- Holmes GL. Cognitive impairment in epilepsy: the role of network abnormalities. *Epileptic Disord*. 2015;17(2):101-116. doi:10.1684/epd.2015.0739
- Kwon JY, Jeon M-T, Jung UJ, Kim DW, Moon GJ, Kim SR. Perspective: therapeutic potential of flavonoids as alternative medicines in epilepsy. *Adv Nutr*. 2019;10(5):778-790. doi:10.1093/advances/nmz047
- Lim H, Heo MY, Kim HP. Flavonoids: broad spectrum agents on chronic inflammation. *Biomol Ther*. 2019;27(3):241-253. doi:10.4062/biomolther.2019.034
- Stompor M. A review on sources and pharmacological aspects of sakuranetin. *Nutrients*. 2020;12(513):1-14. doi:10.3390/nu12020513
- Hasegawa M, Mitsuhara I, Seo S, et al. Analysis on blast fungus-responsive characters of a flavonoid phytoalexin sakuranetin; accumulation in infected rice leaves, antifungal activity and detoxification by fungus. *Molecules*. 2014;19(8):11404-11418. doi:10.3390/molecules190811404
- de Oliveira Melo MN, Oliveira AP, Wicikowski AF, et al. Phenolic compounds from *Viscum album* tinctures enhanced antitumor activity in melanoma murine cancer cells. *Saudi Pharm J*. 2018;26(3):311-322. doi:10.1016/j.jsps.2018.01.011
- Grecco SDS, Reimão JQ, Tempone AG, et al. *In vitro* antileishmanial and antitrypanosomal activities of flavanones from *Baccharis retusa* DC. (Asteraceae). *Exp Parasitol*. 2012;130(2):141-145.
- Danelutte AP, Lago JHG, Young MCM, Kato MJ. Antifungal flavanones and prenylated hydroquinones from *Piper crassinervium* Kunth. *Phytochemistry*. 2003;64(2):555-559. doi:10.1016/S0031-9422(03)00299-1
- Pacciaroni ADELV, Al ET, Pacciaroni AV, et al. Antifungal activity of *Heterothalamus alienus* metabolites. *Phyther Res*. 2008;528(September 2007):524-528. doi:10.1002/ptr.2380
- Aires A, Dias C, Carvalho R, Saavedra MJ. Analysis of glycosylated flavonoids extracted from sweet-cherry stems, as antibacterial agents against pathogenic *Escherichia coli* isolates. *Acta Biochim Pol*. 2017;64(2):265-271. doi:10.18388/abp.2016_1374
- Charles C, Nachtergaeel A, Ouedraogo M, Belayew A, Duez P. Effects of chemopreventive natural products on non-homologous end-joining DNA double-strand break repair. *Mutat*

- Res - Genet Toxicol Environ Mutagen. 2014;768:33-41. doi: [10.1016/j.mrgentox.2014.04.014](https://doi.org/10.1016/j.mrgentox.2014.04.014)
30. Kim KY, Kang H. Sakuranetin inhibits inflammatory enzyme, cytokine, and costimulatory molecule expression in macrophages through modulation of JNK, p38, and STAT1. *Evidence-based Complement Altern Med*. 2016;2016:1-8. doi: [10.1155/2016/9824203](https://doi.org/10.1155/2016/9824203)
 31. Taguchi L, Pinheiro NM, Olivo CR, et al. A flavanone from *Baccharis retusa* (Asteraceae) prevents elastase-induced emphysema in mice by regulating NF- κ B, oxidative stress and metalloproteinases. *Respir Res*. 2015;16(1):1-15. doi: [10.1186/s12931-015-0233-3](https://doi.org/10.1186/s12931-015-0233-3)
 32. Toledo AC, Sakoda CPP, Perini A, Pinheiro NM, Magalhães RM. Flavonone treatment reverses airway inflammation and remodelling in an asthma. *Br J Pharmacol*. 2013;168(7):1736-1749. doi: [10.1111/bph.12062](https://doi.org/10.1111/bph.12062)
 33. Santana FPR, Silva RC, Grecco SS, et al. Inhibition of MAPK and STAT3-SOCS3 by sakuranetin attenuated chronic allergic airway inflammation in mice. *Mediators Inflamm*. 2019;2019:1-14. doi: [10.1155/2019/1356356](https://doi.org/10.1155/2019/1356356)
 34. Terrone G, Frigerio F, Balosso S, Ravizza T, Vezzani A. Inflammation and reactive oxygen species in status epilepticus: biomarkers and implications for therapy. *Epilepsy Behav*. 2019;101(Pt B):1-9. doi: [10.1016/j.yebeh.2019.04.028](https://doi.org/10.1016/j.yebeh.2019.04.028)
 35. Grecco SDS, Doriqueto AC, Landre IM, et al. Structural crystal-line characterization of sakuranetin—an antimicrobial flavanone from twigs of *Baccharis retusa* (Asteraceae). *Molecules*. 2014;19(6):7528-7542. doi: [10.3390/molecules19067528](https://doi.org/10.3390/molecules19067528)
 36. Paxinos G, Franklin K. *Paxinos and Franklin's the Mouse Brain in Stereotaxic Coordinates*. 5th ed. Elsevier; 2004.
 37. Kóvacs KJ. c-Fos as a transcription factor: a stressful re(view) from a functional map. *Neurochem Int*. 1998;33(4):287-297. doi: [10.1016/S0197-0186\(98\)00023-0](https://doi.org/10.1016/S0197-0186(98)00023-0)
 38. Racine RJ. Modification of seizure activity by electrical stimulation: II. Motor seizure. *Electroencephalogr Clin Neurophysiol*. 1972;32(3):281-294. doi: [10.1016/0013-4694\(72\)90177-0](https://doi.org/10.1016/0013-4694(72)90177-0)
 39. Nôga DAMF, Cagni FC, Santos JR, et al. Pro- and anticonvulsant effects of the ant *Dinoponera quadricaps* (Kempf) venom in mice. *Neotrop Entomol*. 2015;44(4):410-417. doi: [10.1007/s13744-015-0292-7](https://doi.org/10.1007/s13744-015-0292-7)
 40. Zhong T, Qing Q, Yang Y, et al. Repression of contextual fear memory induced by isoflurane is accompanied by reduction in histone acetylation and rescued by sodium butyrate. *Br J Anaesth*. 2014;113(4):634-643. doi: [10.1093/bja/aeu184](https://doi.org/10.1093/bja/aeu184)
 41. Ko YH, Kwon SH, Ma SX, et al. The memory-enhancing effects of 7,8,4'-trihydroxyisoflavone, a major metabolite of daidzein, are associated with activation of the cholinergic system and BDNF signaling pathway in mice. *Brain Res Bull*. 2018;142:197-206. doi: [10.1016/j.brainresbull.2018.07.012](https://doi.org/10.1016/j.brainresbull.2018.07.012)
 42. Chen ZM, Fu JF, Shu Q, et al. Diagnosis and treatment recommendations for pediatric respiratory infection caused by the 2019 novel coronavirus. *World J Pediatr*. 2020;16(3):240-246. doi: [10.1007/s12519-020-00345-5](https://doi.org/10.1007/s12519-020-00345-5)
 43. Kanter M, Aktöz T, Aktas C, Ozen F, Yarali O, Kanter B. Role of quercetin in cadmium-induced oxidative stress, testicular damage, and apoptosis in rats. *Anal Quant Cytol Histol*. 2016;38(1):45-51.
 44. Zhong J, Li G, Xu H, Wang Y, Shi M. Baicalin ameliorates chronic mild stress-induced depression-like behaviors in mice and attenuates inflammatory cytokines and oxidative stress. *Braz J Med Biol Res*. 2019;52(7):1-10. doi: [10.1590/1414-431x20198434](https://doi.org/10.1590/1414-431x20198434)
 45. Ai J, Dekermendjian K, Wang X, Nielsen M, Witt M. 6-Methylflavone, a benzodiazepine receptor ligand with antagonistic properties on rat brain and human recombinant GABA_A receptors in vitro. *Drug Dev Res*. 1997;106:99-106. doi: [10.1002/\(SICI\)1098-2299\(199706\)41:2%3C99::AID-DDR7%3E3.0.CO;2-M](https://doi.org/10.1002/(SICI)1098-2299(199706)41:2%3C99::AID-DDR7%3E3.0.CO;2-M)
 46. Jäger AK, Saaby L. Flavonoids and the CNS. *Molecules*. 2011;16(2):1471-1485. doi: [10.3390/molecules16021471](https://doi.org/10.3390/molecules16021471)
 47. Kobayashi S, Nagai T, Konishi Y, Tanabe S, Morimoto K, Ogihara T. Transport mechanisms of flavanone aglycones across Caco-2 cell monolayers and artificial PAMPA membranes. *J Pharm Pharmacol*. 2012;64(1):52-60. doi: [10.1111/j.2042-7158.2011.01374.x](https://doi.org/10.1111/j.2042-7158.2011.01374.x)
 48. Yael D, Tahary O, Gurovich B, Belevovsky K, Bar-Gad I. Disinhibition of the nucleus accumbens leads to macro-scale hyperactivity consisting of micro-scale behavioral segments induced by striatal activity. *J Neurosci*. 2019;39(30):5897-5909. doi: [10.1523/JNEUROSCI.3120-18.2019](https://doi.org/10.1523/JNEUROSCI.3120-18.2019)
 49. Chtourou Y, Slima AB, Gdoura R, Fetoui H. Naringenin mitigates iron-induced anxiety-like behavioral impairment, mitochondrial dysfunctions, ectonucleotidases and acetylcholinesterase alteration activities in rat hippocampus. *Neurochem Res*. 2015;40(8):1563-1575. doi: [10.1007/s11064-015-1627-9](https://doi.org/10.1007/s11064-015-1627-9)
 50. Gadotti VM, Zamponi GW. Anxiolytic effects of the flavonoid luteolin in a mouse model of acute colitis. *Mol Brain*. 2019;12(1):10-12.
 51. Germán-Ponciano LJ, Puga-Olguín A, De Jesús Rovirosa-Hernández M, Caba M, Meza E, Rodríguez-Landa JF. Differential effects of acute and chronic treatment with the flavonoid chrysin on anxiety-like behavior and Fos immunoreactivity in the lateral septal nucleus in rats. *Acta Pharm*. 2020;70(3):387-397. doi: [10.2478/acph-2020-0022](https://doi.org/10.2478/acph-2020-0022)
 52. Samad N, Saleem A, Yasmin F, Shehzad MA. Quercetin protects against stress-induced anxiety- and depression- like behavior and improves memory in male mice. *Physiol Res*. 2018;67(5):795-808. doi: [10.33549/physiolres.933776](https://doi.org/10.33549/physiolres.933776)
 53. Hinton T, Johnston GAR. GABA, the major inhibitory neurotransmitter in the brain. In: *Reference Module in Biomedical Sciences*. 2nd ed. Elsevier Inc.; 2018:1-15.
 54. Leonard B. Drug treatment of the epilepsies. *Fundam Psychopharmacol*. 2004;295-318.
 55. Saft C, Speckmann E-J. Antiepileptic effects of cobalt, manganese and magnesium on bicuculline-induced epileptiform activity in hippocampal neurons. *Brain Res*. 2020;1732:1-8. doi: [10.1016/j.brainres.2020.146684](https://doi.org/10.1016/j.brainres.2020.146684)
 56. Meldrum B. GABAergic mechanisms in the pathogenesis and treatment of epilepsy. *Br J Clin Pharmacol*. 1989;27(1 S):3S-11S.
 57. Bernaskova K, Mares P. Similar effects of lamotrigine and phenytoin against cortical epileptic foci in immature rats. *Physiol Res*. 2010;59(1):113-119.
 58. Van der Linde HJ, Van Deuren B, Somers Y, Teisman A, Drinkenburg WH, Gallacher DJ. EEG in the FEAB model: measurement of electroencephalographical burst suppression and seizure liability in safety pharmacology. *J Pharmacol Toxicol Methods*. 2011;63(1):96-101. doi: [10.1016/j.vascn.2010.06.002](https://doi.org/10.1016/j.vascn.2010.06.002)
 59. Capasso A, Gallo C. Anticonvulsant activity of new GABA pro-drugs. *Med Chem (Los Angeles)*. 2012;5(4):343-352.
 60. Nkantchoua GCN, Njapdounke JSK, Fifen JJ, et al. Anticonvulsant effects of *Senna spectabilis* on seizures induced by chemicals and maximal electroshock. *J Ethnopharmacol*. 2018;212:18-28. doi: [10.1016/j.jep.2017.09.042](https://doi.org/10.1016/j.jep.2017.09.042)
 61. Faggion SA, Cunha AOS, Fachim HA, et al. Anticonvulsant profile of the alkaloids (+)-erythravine and (+)-11- α -hydroxy-erythravine isolated from the flowers of *Erythrina mulungu* Mart ex Benth (Leguminosae-Papilionaceae). *Epilepsy Behav*. 2011;20(3):441-446. doi: [10.1016/j.yebeh.2010.12.037](https://doi.org/10.1016/j.yebeh.2010.12.037)
 62. Leclercq K, Matagne A, Provins L, Klitgaard H, Kaminski RM. Pharmacological profile of the novel antiepileptic drug candidate padsevonil: characterization in rodent seizure and epilepsy models. *J Pharmacol Exp Ther*. 2020;372(1):11-20. doi: [10.1124/jpet.119.261222](https://doi.org/10.1124/jpet.119.261222)
 63. Mussi-Ribeiro A, Miranda A, Gobbo-Netto L, Pepporine Lopes N, Dos Santos WF. A anticonvulsive fraction from *Scaptocosa*

- raptoria* (Araneae: Lycosidae) spider venom. *Neurosci Lett*. 2004;371(2-3):171-175. doi:[10.1016/j.neulet.2004.08.064](https://doi.org/10.1016/j.neulet.2004.08.064)
64. Nôga DAMF, Brandão LEM, Cagni FC, et al. Anticonvulsant effects of fractions isolated from *Dinoponera quadricaps* (Kempt) ant venom (Formicidae: Ponerinae). *Toxins (Basel)*. 2017;9(1):1-13. doi:[10.3390/toxins9010005](https://doi.org/10.3390/toxins9010005)
 65. Taiwe GS, Tchaya TB, Menanga JR, Dabole B, De Waard M. Anticonvulsant activity of an active fraction extracted from *Crinum jagus* L. (Amaryllidaceae), and its possible effects on fully kindled seizures, depression-like behaviour and oxidative stress in experimental rodent models. *J Ethnopharmacol*. 2016;194:421-433. doi:[10.1016/j.jep.2016.10.023](https://doi.org/10.1016/j.jep.2016.10.023)
 66. Addae J, Pingal R, Walkins K, Cruickshank R, Youssef F, Nayak S. Effects of *Jasminum multiflorum* leaf extract on rodent models of epilepsy, motor coordination and anxiety. *Epilepsy Res*. 2017;131:58-63. doi:[10.1016/j.epilepsyres.2017.02.012](https://doi.org/10.1016/j.epilepsyres.2017.02.012)
 67. Gawande D, Druzhilovsky D, Gupta RC, Poroikov V, Goel RK. Anticonvulsant activity and acute neurotoxic profile of *Achyranthes aspera* Linn. *J Ethnopharmacol*. 2017;202:97-102. doi:[10.1016/j.jep.2017.03.018](https://doi.org/10.1016/j.jep.2017.03.018)
 68. Taiwe GS, Kouamou ALN, Ambassa ARM, Menanga JR, Tchaya TB, Dzeufiet PDD. Evidence for the involvement of the GABA-ergic pathway in the anticonvulsant activity of the roots bark aqueous extract of *Anthocleista djalensis* A. Chev (Loganiaceae) *J Basic Clin Physiol Pharmacol*. 2017;28(5):425-435. doi:[10.1515/jbcp-2017-0048](https://doi.org/10.1515/jbcp-2017-0048)
 69. Luszczki JJ, Swiader MJ, Swiader K. Anticonvulsant and acute adverse effect profiles of picolinic acid 2-fluoro-benzylamide in various experimental seizure models and chimney test in mice. *Fundam Clin Pharmacol*. 2008;22(1):69-74. doi:[10.1111/j.1472-8206.2007.00547.x](https://doi.org/10.1111/j.1472-8206.2007.00547.x)
 70. Keller SS, Richardson MP, O'Muircheartaigh J, Schoene-Bake JC, Elger C, Weber B. Morphometric MRI alterations and postoperative seizure control in refractory temporal lobe epilepsy. *Hum Brain Mapp*. 2015;36(5):1637-1647. doi:[10.1002/hbm.22722](https://doi.org/10.1002/hbm.22722)
 71. Wu D, Chang F, Peng D, Xie S, Li X, Zheng W. The morphological characteristics of hippocampus and thalamus in mesial temporal lobe epilepsy. *BMC Neurol*. 2020;20(235):1-9. doi:[10.1186/s12883-020-01817-x](https://doi.org/10.1186/s12883-020-01817-x)
 72. Kuba R, Tyrliková I, Pažourková M, et al. Grey-white matter abnormalities in temporal lobe epilepsy associated with hippocampal sclerosis: inter-observer analysis, histopathological findings, and correlation with clinical variables. *Epilepsy Res*. 2012;102(1-2):78-85. doi:[10.1016/j.epilepsyres.2012.05.002](https://doi.org/10.1016/j.epilepsyres.2012.05.002)
 73. Jardim AP, Corso JT, Garcia MTFC, et al. Hippocampal atrophy on MRI is predictive of histopathological patterns and surgical prognosis in mesial temporal lobe epilepsy with hippocampal sclerosis. *Epilepsy Res*. 2016;128:169-175. doi:[10.1016/j.epilepsyres.2016.10.014](https://doi.org/10.1016/j.epilepsyres.2016.10.014)
 74. Hawley DF, Morch K, Christie BR, Leasure JL. Differential response of hippocampal subregions to stress and learning. *PLoS ONE*. 2012;7(12):1-7. doi:[10.1371/journal.pone.0053126](https://doi.org/10.1371/journal.pone.0053126)
 75. Amaral DG, Scharfman HE, Lavenex P. The dentate gyrus: fundamental neuroanatomical organization (dentate gyrus for dummies). *Prog Brain Res*. 2008;163:3-22. doi:[10.1016/S0079-6123\(07\)63001-5](https://doi.org/10.1016/S0079-6123(07)63001-5)
 76. Scharfman HE. *The Dentate Gyrus: A Comprehensive Guide to Structure, Function, and Clinical Implications*. Elsevier; 2007.
 77. Dengler CG, Coulter DA. Normal and epilepsy-associated pathologic function of the dentate gyrus. *Prog Brain Res*. 2016;226:155-178. doi:[10.1016/bs.pbr.2016.04.005](https://doi.org/10.1016/bs.pbr.2016.04.005)
 78. Scharfman HE, MacLusky NJ. Differential regulation of BDNF, synaptic plasticity and sprouting in the hippocampal mossy fiber pathway of male and female rats. *Neuropharmacology*. 2014;76(PART C):696-708.
 79. Botterill JJ, Lu YL, LaFrancois JJ, et al. An excitatory and epileptogenic effect of dentate gyrus mossy cells in a mouse model of epilepsy. *Cell Rep*. 2019;29(9):2875-2889.e6. doi:[10.1016/j.celrep.2019.10.100](https://doi.org/10.1016/j.celrep.2019.10.100)
 80. Kahn JB, Port RG, Yue C, Takano H, Coulter DA. Circuit-based interventions in the dentate gyrus rescue epilepsy-associated cognitive dysfunction. *Brain*. 2019;142(9):2705-2721. doi:[10.1093/brain/awz209](https://doi.org/10.1093/brain/awz209)
 81. Marder M, Paladini A. GABA(A)-receptor ligands of flavonoid structure. *Curr Top Med Chem*. 2002;2(8):853-867. doi:[10.2174/1568026023393462](https://doi.org/10.2174/1568026023393462)
 82. Johnston GAR, Beart PM. Flavonoids: some of the wisdom of sage? *Br J Pharmacol*. 2004;142(5):809-810. doi:[10.1038/sj.bjp.0705827](https://doi.org/10.1038/sj.bjp.0705827)

How to cite this article: Vicente-Silva W, Silva-Freitas FR, Beserra-Filho JIA, et al. Sakuranetin exerts anticonvulsant effect in bicuculline-induced seizures. *Fundam Clin Pharmacol*. 2022;36(4):663-673. doi:[10.1111/fcp.12768](https://doi.org/10.1111/fcp.12768)